

Ball Pit Brawl — Game Design Spec

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Overview

Ball Pit Brawl is a foam-weapon shooter set inside a giant multistory indoor play center: players climb nets, ride tube slides, and wade through ball pits while fighting with foam knives, foam guns, and plastic balls that knock a player out in one hit.

Core loop: spawn with a foam knife and a single-shot foam gun → climb toward the enemy → earn score from kills and objectives → spend score on better foam guns and darts → fight over the ball cannons, the most powerful weapons in the arena.

Design pillars

- **The playground is the battlefield.** Every piece of play equipment is also a route or a piece of cover: nets you can see through but not shoot through, slides that are fast but trap you, ball pits that hide you but slow you down.
- **Balls are deadly, darts are reliable.** Any live ball is a one-hit knockout, so balls are the high-risk, high-reward weapon. Darts do steady damage and never run out if you keep vacuuming.
- **One shared ball supply.** Every ball in the arena cycles through one conveyor system that feeds every cannon. Grabbing balls to throw, or draining the conveyor by firing a cannon, takes them away from everyone else.
- **Earn your arsenal.** Everyone starts equal; score buys better guns mid-match.

Camera: first person. Players see their foam gun or knife in hand, and climbing, sliding, and wading through ball pits are all shown from the player's eyes.

Setting and arena

The whole match takes place inside one huge indoor play structure, a steel frame of padded decks, nets, tube slides, and ball pits four levels high, inside a warehouse-sized building.



The first image sets the look of each structure (colored pipe frame, padded floors, spiral and tube slides, a ball pit at the base). The second sets the scale and the open atrium: wide drops between netted decks, ball pits below, and the overhead pipes that inspired the ball conveyor.

Layout (default map, 60 m × 30 m hall)

- Two team towers, one at each end of the hall, each holding that team's base, spawn, and store. In free for all, both stores are open to everyone and spawns are spread across the map.
- A central atrium between the towers, open from floor to ceiling, crossed by rope bridges, net tunnels, and tube slides. This is the long-range danger zone.
- Mirror-symmetric, so neither team has a better side.

Level	Height	What's there
Ground	0 m	Large ball pits, the main conveyor belts, crawl tunnels, slide exits
Level 2	3 m	Padded decks, obstacle courses (punching bags, rollers), base entrances
Level 3	6 m	Rope bridges across the atrium, net tunnels, team cannons
Roof deck	9 m	Spiral slide tops, two neutral cannons at the center, highest sight lines

Cover materials

Material	Blocks darts	Blocks balls	Can see through	Notes
Solid plastic panel	Yes	Yes	No	Full cover
Netted wall	Yes	Yes	Yes	You can see enemies but not shoot through; push around the edges to get a shot
Open frame	No	No	Yes	Pipe railings only; no protection

Material	Blocks darts	Blocks balls	Can see through	Notes
Foam padding blocks	Yes	Yes	No	Low cover on decks; crouch behind
Tube slide shell	Yes	Yes	Partly (colored windows)	Fast travel, but you're trapped inside until the exit

Netted walls are the main way players cross the big open spaces safely: routes along the atrium edge are netted, while the fastest routes (bridges, open decks) are exposed.

Movement and traversal

Getting to the enemy means climbing up and sliding down: the fast routes are exposed or one-way, and the safe routes are slow.

Traversal	Speed	Can attack?	Trade-off
Run on decks	5.5 m/s	Yes	Default
Crouch	2.5 m/s	Yes	Hides behind padding blocks
Jump	1.0 m high	Yes	Hop gaps and low walls
Climb cargo net or ladder	1.5 m/s	No	Main way up a level; you're exposed
Crawl tunnel	2.0 m/s	Knife only	Fully enclosed, safe but slow
Tube or spiral slide	8 to 10 m/s	No	Fastest way down; can't stop or exit until the bottom
Rope bridge	4.0 m/s	Yes	Crosses the atrium; wobbles aim (+3° spread)
Wade through ball pit	2.0 m/s	Yes	Free balls to grab everywhere
Submerge in ball pit	1.0 m/s	No	Hidden from view and immune to darts; can't see out except a narrow peek
Drop down from a ledge	Gravity	Yes	Falls up to 3 m are safe; landing in a ball pit is safe from any height

Players have 100 health and respawn after 5 s (except in Disarm the bomb, which has no respawns within a round).

Weapons

Every player starts with a foam knife and a single-shot foam gun; better guns are bought with score, and any live ball knocks a player out in one hit.

Damage model

- Players have 100 health. Health regenerates after 4 s without taking damage.
- Foam darts are slow (15 to 30 m/s) and drop over distance, so long shots need leading.
- Balls are faster and flatter than darts (25 m/s thrown, 40 m/s from a cannon) and are an instant knockout.

- A ball is only deadly until its first bounce, like dodgeball. After that it's just a ball on the floor (and heads for the conveyor).
- Friendly fire is off in team modes.

Foam knife (always carried)

- 50 damage per swing, 0.6 s between swings, 2 m reach. Two hits knock out a full-health player.
- A hit from behind is an instant knockout.
- The only attack that works inside crawl tunnels.

Thrown balls

- Pick up a ball by walking over it or reaching into a ball pit (0.4 s). You can carry one ball at a time.
- Holding a ball, your gun is lowered. Throwing has a 0.3 s windup, then the ball flies at 25 m/s with light drop.
- Any player can grab a ball anywhere: ball pits are full of them, which is what makes the ground floor dangerous.

Foam guns (bought in the store)

Gun	Cost (score)	Damage per dart	Magazine	Fire rate	Dart speed	Role
Single-shot blaster	Free (starter)	35	1	Reload 1.0 s	20 m/s	Starter; 3 hits to knock out
Semi-auto dart pistol	150	25	8	3 shots/s	20 m/s	Cheap first upgrade
Six-shot revolver	300	30	6	2 shots/s	20 m/s	Reliable mid-range sidearm
Dual mini blasters	450	20	2 × 4	4 shots/s (alternating)	18 m/s	Fast close-range spray; no aiming down sights
Pump scatter blaster	600	12 × 6 darts	4 shells	1 shot/s	15 m/s	Close-range bursts in tunnels and on decks
Three-round burst blaster	750	20	18	3-dart bursts, 2 bursts/s	22 m/s	Accurate mid-range
Flywheel auto blaster	900	14	25	8 shots/s	22 m/s	Sustained fire, eats ammo fast

Gun	Cost (score)	Damage per dart	Magazine	Fire rate	Dart speed	Role
Mega dart crossbow	1,000	90	1	Reload 1.5 s	18 m/s	One huge dart with heavy drop; knocks the target back 1 m
Belt-fed heavy blaster	1,200	12	100	10 shots/s after 1.5 s spin-up	20 m/s	Holds a doorway; slows you to 80% speed
Long-range blaster	1,300	75	5	1 shot/s	30 m/s	Across the atrium; low dart drop
Foam rocket launcher	1,600	60, splash within 2 m	1	Reload 3.0 s	14 m/s	Clears decks and campers; slow rocket is easy to dodge

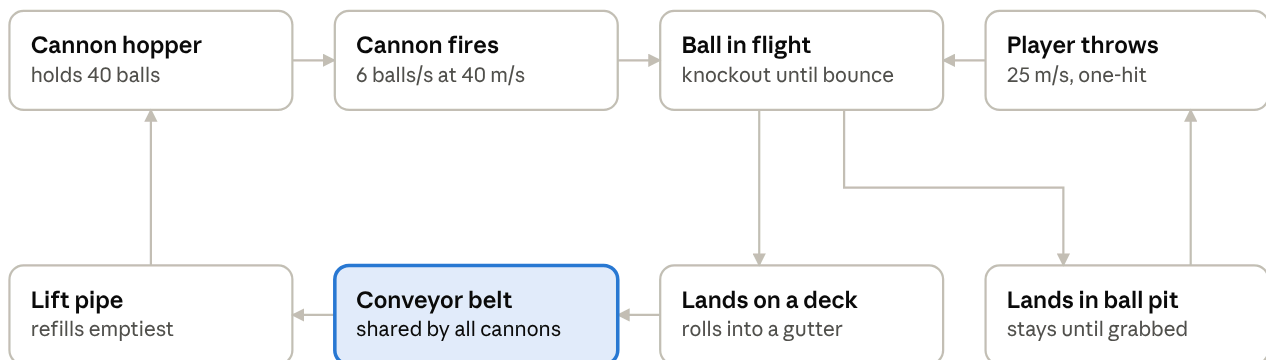
The launcher's rockets use the same vacuum-collectable ammo as darts but cost 5 darts each.

Players carry the starter blaster, one bought gun, and the knife. Buying a new gun replaces the old bought gun. Bought guns are kept after being knocked out, for the rest of the match.

Ball cannons and the conveyor loop

Ball cannons are the fastest-firing weapons in the game, but they are bolted in place and can only fire balls that the shared conveyor brings back to them.

Every ball cycles back to the cannons through one shared conveyor



ball loop · cannons, conveyor, pits

Balls that land on decks or floors roll into gutters and ride the conveyor back to the cannons; balls that land in a ball pit stay there until a player grabs one to throw.

Cannons

- Six cannons: two in each team tower on Level 3 aimed across the atrium, and two neutral cannons on the central roof deck. Any player can use any cannon.
- Fire rate 6 balls per second at 40 m/s, nearly flat. Every ball is a one-hit knockout until its first bounce.
- Turning limits: 120° left to right and 40° up and down, so each cannon has blind spots players can use.
- Getting on takes 0.5 s; getting off is instant. The operator sits exposed from behind and above.
- Overheats after 3 s of continuous fire and needs 2 s to cool down.

The conveyor

- Every deck and floor slopes slightly toward gutters at its edges. Gutters drop onto conveyor belts running along the ground floor.
- The belts carry balls to a hub, where air pipes (like the overhead pipes in the reference image) lift them up to the cannon hoppers. The lift always refills the hopper with the fewest balls first.
- A ball takes about 8 s to get from a gutter to a hopper. Each hopper holds 40 balls and starts the match full.
- Balls dropped by a knocked-out player, and missed throws that land on decks, also go back into the conveyor.
- One shared supply means a team firing a cannon nonstop drains the balls everyone else's cannons depend on, and a team that keeps balls in its hands is starving the cannons.

Score, store, and ammo

Score is both the leaderboard and the currency: you earn it by knocking out enemies and playing the objective, and spend it at your team's store on guns and darts. Spending doesn't lower your leaderboard total.

Action	Score earned
Knockout with knife, dart, or thrown ball	100

Action	Score earned
Knockout with a cannon	50 (cannons are easy, so they pay less)
Assist (damaged a player a teammate knocked out within 5 s)	50
Capture the flag	300
Return your flag	100
Plant the bomb	200
Disarm the bomb	300

The store

- One store in each team's tower (both open to everyone in free for all). It's a counter you walk up to; buying is instant from a quick menu.
- Sells the foam guns in the weapons table, plus darts: 12 darts for 50 score.
- The store area is a safe zone for 3 s after you arrive, so players aren't knocked out mid-menu.

Vacuum reloading (free)

- Every player has a handheld vacuum. Hold the vacuum button to suck up foam darts lying within a 4 m cone in front of you, at 10 darts per second.
- You can keep moving at 70% speed while vacuuming, but can't fire. Release to fire immediately.
- Fired darts stay on the floor for 60 s. Enemy darts work too, so fights leave free ammo behind for the winner.
- The vacuum only collects darts. Balls are picked up by hand or carried by the conveyor.
- Dart reserve: start with 12, carry up to 60.

Joke shop

Every store has a joke shelf of very cheap items that change nothing about the fight. They exist to make teammates laugh.

Item	Cost (score)	What it does
Party horn	5	One loud toot and a little noise-maker animation

Item	Cost (score)	What it does
Juice box	5	You drink it with an extremely long slurp
Dad joke card	5	Shows a random terrible joke on your teammates' screens
Kazoo	5	Play a wobbly 3-second tune
Confetti popper	5	A burst of confetti that vanishes in 2 s
Whoopee cushion	10	Place it on a deck; the next teammate to step on it sets it off
Rubber chicken	10	Throw it; it bounces around squawking and does nothing else
Capsule machine toy	10	A random cheap plastic prize with a silly name, added to your collection
Paper birthday crown	15	Put it on a teammate's head for the rest of their life
Giant foam finger	15	Wave it around for 10 s in place of your weapon
Propeller beanie	20	The propeller spins faster the faster you run

Rules that keep them harmless: no joke item changes hitbox size, speed, damage, or what enemies can see through. Sounds from joke items are heard only by teammates. Each item has a 10 s cooldown so nobody spams it.

Streak rewards

Staying alive pays: score earned without being knocked out builds a streak, and hitting a streak threshold unlocks a reward. There are six rewards, and each player picks three before the match.

How streaks work

- Streak points are the score you've earned since your last knockout (kills, assists, objectives). Being knocked out resets them to zero.
- Spending score in the store does not lower your streak.
- Each of your three rewards unlocks once per life when your streak reaches its threshold. Press the reward key to use it whenever you like; an unused reward is lost if you're knocked out.

- You choose your three rewards in the pre-match loadout screen and can swap them while respawning.

Reward	Streak needed	What it does	How to counter it
Instant dart refill	300	Fills your magazine and dart reserve to max, plus any teammates within 10 m	None; it's a small reward
Placeable health box	400	Drop a first-aid box on a deck. Teammates who touch it heal 50 health. Lasts 60 s or 5 uses	Destroy it with 3 darts or 1 ball
Swarm of annoying kids	600	8 shrieking kids flood the enemy half for 30 s: they run in front of enemies, crowd slides and ladders, and splash around in ball pits	They can't be hurt (darts bounce off and they laugh); wait them out or go around
RC car drone	700	Drive a remote-control car with a dart turret (10 damage, 4 shots/s) through tunnels and decks for 45 s. It has 50 health. Your body stands still while you drive	Shoot the car, or find and knock out the driver
Flying dart drone	900	A drone hovers over you for 30 s and fires at any enemy it can see (8 damage, 3 shots/s)	Shoot it down (80 health) or break line of sight; nets block its darts
Dart tank	1,200	Ride a padded ride-on toy tank for 40 s with twin flywheel blasters (15 damage, 10 shots/s) and 400 armor. Balls do 100 damage to it instead of a knockout	Stay on narrow upper decks it can't reach; focus it with balls

The tank only drives on the ground floor and the wide Level 2 decks. The RC car fits through crawl tunnels.

Game modes

All four modes use the same arena, weapons, and economy; they change the goal, team setup, and respawn rules.

Mode	Players	Goal	Respawn	Ends when
Free for all	8 to 12, solo	Most knockouts	5 s, spread spawns	First to 25 knockouts, or 10 min
Team deathmatch	2 teams of 4 to 6	Team knockouts	5 s, in your tower	First team to 50, or 10 min
Capture the flag	2 teams of 4 to 6	Take the enemy flag to your base	8 s, in your tower	First to 3 captures, or 15 min
Disarm the bomb	2 teams of 5 (attack vs. defend)	Attackers plant, defenders stop or disarm	None within a round	First to 7 rounds; sides swap at round 6

Capture the flag details

- Each flag sits on the roof deck of its team's tower, so attackers must climb all the way up.
- The carrier can use the knife and starter blaster only (no ball throwing, no bought gun), and moves at 90% speed.
- Carriers can ride slides. Sliding down with the flag is the fastest escape, but the slide exit is a known spot to wait at.
- A dropped flag returns home after 20 s, or immediately when a defender touches it.

Disarm the bomb details

- Each round, the attackers carry one foam "bomb" (a big, obvious prop). There are two plant sites inside the defenders' tower: one on Level 2, one on Level 3.
- Planting takes 4 s; the bomb then goes off after 40 s. Disarming takes 6 s, or 3 s with a disarm kit from the store (200 score).
- Rounds last 2 min before the plant. Attackers win by detonation or by knocking out every defender; defenders win by disarming, running out the clock, or knocking out every attacker before a plant.
- Score carries between rounds, so buying is part of the strategy.

Technical notes and build milestones

Use the same pipeline as Warp Dodgeball: Claude Code models the play structure in Blender, exports .glb files, and the game runs in Godot 4. The biggest technical risk is the ball count, so it's built and tested early.

Engine notes

- **Ball pits:** thousands of real physics balls would be far too slow. Draw resting pit balls as one instanced mesh with a gentle animated surface, and only spawn a real physics ball when a player grabs one, a ball is fired, or a ball drops. Real balls that settle get absorbed back into the pit.
- **Ball supply:** track the total number of balls as one number shared by pits, conveyors, hoppers, and balls in flight, so the shared-supply rule is exact.
- **Darts:** fast projectiles with drop, each a lightweight object. Cap live and floor darts at around 400; remove the oldest first.
- **Nets:** use collision that blocks projectiles but not line of sight (a see-through material on a solid collider).
- **Slides:** move the player along a fixed path (spline) rather than simulating sliding physics.
- **Blender conventions:** one file per modular piece (deck, net wall, slide section, cannon, conveyor segment), snapped to a 1 m grid so Claude Code can assemble maps from pieces. Collision meshes use the `-col` suffix.

Milestones

1. **Greybox modular kit and one small map.** Decks, nets, ladders, a slide, a ball pit.
 - Test: a player can climb from the ground to Level 3 and slide back down.
2. **Movement.** All traversal types from the movement table.
 - Test: speeds match the table; submerged players can't be hit by darts.
3. **Knife and starter blaster, health, respawn.**
 - Test: 2 knife hits or 3 starter darts knock out a full-health player; a back hit with the knife is instant.
4. **Balls.** Pit pickup, throwing, one-hit knockout until first bounce, instanced pits.
 - Test: 60 frames per second holds with 2,000 pit balls and 100 live balls.
5. **Conveyor and cannons.** The full ball loop.
 - Test: firing a cannon until empty, then waiting, refills it only from balls that came back through the conveyor.

6. Score, store, vacuum, gun tiers.

- Test: buying with exactly enough score works; vacuuming picks up both teams' darts.

7. Modes. Team deathmatch first, then free for all, capture the flag, disarm the bomb, with simple bots.

- Test: each mode completes with bots and declares the right winner.

8. Full map and art pass. The 60 m × 30 m arena in Blender, colors and padding to match the reference images, sounds, HUD.

Open questions

Each has a working default in the spec; confirm or change it before the matching milestone.

- ☐ **Cannon balance:** cannons are the fastest-firing weapons and every ball is a one-hit knockout. The spec limits them with a fixed firing arc, an exposed operator, overheating, and the shared ball supply. Is that enough, or should cannon balls do damage instead of instant knockouts?
- ☐ **Ball speed:** the spec makes balls faster than darts, as described, including thrown balls. Should thrown balls be slower than cannon balls but still faster than darts (current default)?
- ☐ **Dodgeball bounce rule:** balls stop being deadly after one bounce. Keep it, or let bounced balls still knock players out?
- ☐ **Players:** bots first, then online? Default: bots first.
- ☐ **Knife role:** the spec makes the knife a melee weapon with backstab knockouts. Did you mean something more specific by knives being for attacking the enemy base?
- ☐ **Streak thresholds:** should streaks count only knockouts instead of all score? Counting objective score (default) rewards flag runners and bomb planters too.